The empirical recurrence for OEIS A264208, recovered from its tail

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8 September 2026

Abstract

OEIS A264208 records "Empirical recurrence of order 92" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 4096$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the whole sequence, so the recurrence holds from $n = 93$ — every index at which it can be written down. Its last coefficient is $c_{92} = 1 \neq 0$, so no further tail satisfies anything shorter; any order-92 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A264208 is "Number of (2+1)X(n+1) arrays of permutations of 0..n*3+2 with each element having index change +-(.,.) 0,0 0,2 or 1,0.". It has offset 1 and begins

$$9, 96, 1024, 7328, 52441, 422505, 3404025, 26420400, \ldots$$

The entry states:

> Empirical recurrence of order 92 (see link above)

As of the "Last modified" line on the live entry (Nov 10 2015 11:38:03, revision 8) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 4096$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 4096$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Here the stated order is already the minimal order of the sequence itself, so no tail has to be taken.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 8192$ exact terms from $n = 1$ on, it returns order 92 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{92} c_i \, a(n-i),$$

c_1	6	c_{24}	−40020183381	c_{47}	76092524885584	c_{70}	65442067712
c_2	4	c_{25}	319100874902	c_{48}	185968952128093	c_{71}	−28497798632
c_3	60	c_{26}	800489607316	c_{49}	−97173867352798	c_{72}	−7104902471
c_4	683	c_{27}	−592673531516	c_{50}	−54280458193908	c_{73}	−391920574
c_5	−3136	c_{28}	−283712439929	c_{51}	−23049051895420	c_{74}	−2983585700
c_6	−3484	c_{29}	−1981131271080	c_{52}	−105438675879839	c_{75}	1416741948
c_7	−21386	c_{30}	−5647506286604	c_{53}	64339708655248	c_{76}	432442925
c_8	−144225	c_{31}	5052145952710	c_{54}	51817107999788	c_{77}	−45213224
c_9	507964	c_{32}	5452725674571	c_{55}	−5385080044430	c_{78}	70567548
c_{10}	778304	c_{33}	6242062101508	c_{56}	33413928974141	c_{79}	−37747374
c_{11}	1828540	c_{34}	23013931256128	c_{57}	−23870665585532	c_{80}	−11907687
c_{12}	11907687	c_{35}	−23870665585532	c_{58}	−23013931256128	c_{81}	1828540
c_{13}	−37747374	c_{36}	−33413928974141	c_{59}	6242062101508	c_{82}	−778304
c_{14}	−70567548	c_{37}	−5385080044430	c_{60}	−5452725674571	c_{83}	507964
c_{15}	−45213224	c_{38}	−51817107999788	c_{61}	5052145952710	c_{84}	144225
c_{16}	−432442925	c_{39}	64339708655248	c_{62}	5647506286604	c_{85}	−21386
c_{17}	1416741948	c_{40}	105438675879839	c_{63}	−1981131271080	c_{86}	3484
c_{18}	2983585700	c_{41}	−23049051895420	c_{64}	283712439929	c_{87}	−3136
c_{19}	−391920574	c_{42}	54280458193908	c_{65}	−592673531516	c_{88}	−683
c_{20}	7104902471	c_{43}	−97173867352798	c_{66}	−800489607316	c_{89}	60
c_{21}	−28497798632	c_{44}	−185968952128093	c_{67}	319100874902	c_{90}	−4
c_{22}	−65442067712	c_{45}	76092524885584	c_{68}	40020183381	c_{91}	6
c_{23}	33864208408	c_{46}	0	c_{69}	33864208408	c_{92}	1

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 93$, and it is the recurrence OEIS A264208 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 1$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{92} - c_1 x^{92-1} - \cdots - c_{92}$. Its constant term is $-c_{92} = -1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 92 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 92, so $\deg q = 92$, and it is monic. It holds from some index t on. From $\max(t, 1)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 92, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 92, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-92 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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